

2(a)	$T \sin 68^\circ + 32 = 280$	C1
	$T = 270 \text{ N}$	A1
2(b)(i)	$F = \rho g V$ $V = 280 / (1.0 \times 10^3 \times 9.81)$ $= 0.029 \text{ m}^3$	A1
2(b)(ii)	$\rho = (32 / 9.81) / 0.029$	C1
	$= 110 \text{ kg m}^{-3}$	A1

Question	Answer	Marks
2(c)	$(\Delta)E = mg(\Delta)h$ or $(\Delta)E = W(\Delta)h$	C1
	$(\Delta)h = (-) 77 / 32$	C1
	$(\Delta)h = (-) 2.4$ final height = $6.2 - 2.4$ $= 3.8 \text{ m}$	A1
2(d)(i)	$T = kx$ where k is a constant or $T = (EA / L)x$ where A is (cross-sectional) area, E is Young modulus, L is (original/unstretched) length	B1
2(d)(ii)	$E = \frac{1}{2} kx^2$ or $E = \frac{1}{2} Fx$ and $F = kx$	C1
	$E = 0.65 \times 2^2$ $= 2.6 \text{ J}$	A1
	or	
	$E = \frac{1}{2} Fx$ $0.65 = \frac{1}{2} \times 270 \times x$ and so $x = 4.8 \times 10^{-3} \text{ m}$ $k = F / x = 270 / 4.8 \times 10^{-3}$ $= 5.6 \times 10^4$	(C1)

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Question	Answer	Marks
2(d)(ii)	final $E = \frac{1}{2} \times 540 \times 9.6 \times 10^{-3}$ or $E = \frac{1}{2} \times (270 \times 2) \times (4.8 \times 10^{-3} \times 2)$ or $E = \frac{1}{2} \times 5.6 \times 10^4 \times (9.6 \times 10^{-3})^2$ $= 2.6 \text{ J}$	(A1)